

Preregistration for: “High seas pocket closures”

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1 Data collection

Have any data been collected for this study already?

It’s complicated (See Q8 for details)

2 Hypotheses

What’s the main question being asked or hypothesis being tested in this study?

Our study will test three hypotheses:

- H¹: The PNA’s closure of the high-seas pockets to purse seine fishing reduced purse seine fishing effort within the high seas pockets.
- H²: Reducing purse seine fishing effort within the high seas pockets will increase catch-per-unit-effort of tuna, especially Bigeye tuna (*Thunnus obesus*), for the longline fleet operating within the high seas pockets.
- H³: The closure will also increase catch-per-unit-effort of Bigeye (*Thunnus obesus*), Skipjack (*Katsuwonus pelamis*) and Yellowfin tuna (*Thunnus albacares*) to the purse seine and longline fleets operating in PNA waters within 100 nautical miles of the high seas pockets.

3 Dependent variable

Describe the key dependent variable(s) specifying how they will be measured.

For H¹: Total purse seine fishing effort (sets and days) by grid cell and year, as provided by public domain catch and effort data in the WCPFC catalog.

For H²⁻³: Catch-per-unit-effort (CPUE) by grid cell and year: $CPUE_{it} = \frac{\sum_{j \in (i,t)} Catch_j}{\sum_{j \in (i,t)} Effort_j}$, accommodating for gear-specific differences. Longline CPUE will be measured in number of fish per thousand hooks and metric tons per thousand hooks, with 5°x5° grid cells. For purse seines, CPUE is measured in metric tons per set and metric tons per day; grid cells are 1°x1°.

4 Conditions

How many and which conditions will participants be assigned to?

All tests have periods for before / after high seas closure (effective January 2010), as well as treatment and control groups. The definition of treatment and control groups varies by test, as follows:

- For H¹: Treatment grid cells are those within the high seas pockets, and control grid cells are other high seas grid cells that fall under the WCPFC convention area and lie between 20°N–20°S.
- For H²: Grid cells inside high seas pockets are “treated” and other high seas grid cells that fall under the WCPFC convention area and lie between 20°N–20°S will be considered “control”.
- For H³: Grid cells within 100 nautical miles of a high seas pocket boundary are “treated”. Grid cells between 100 and 200 nautical miles of a high seas pocket boundary are “control”. For purse seine, we will only retain grid cells that fall within PNA countries to account for the sharp difference in management between vessels that operate under a vessel-day scheme vs loosely regulated high seas.

In all cases, data from 2009 and before will be labeled as “pre”-closure and data 2010 and after will be labeled as “post”-closure.

5 Analyses

Specify exactly which analyses you will conduct to examine the main question/hypothesis.

For H¹: Let E_{it} be purse seine effort (measured in sets or days) in grid cell i at time t . Then, we will perform a linear regression with the form:

$$E_{it} = \sum_{t \neq 0} \beta_t T_t + \mu_i + \epsilon_{it} \quad (1)$$

Where T_t are indicators for periods relative to year of treatment, such that $T_t = 0$ corresponds to the last year when purse seine fishing was allowed, 2009. Negative values of T_t are pre-closure years, and positive values of T_t are post-closure years. The parameter of interest is β_t , which is a vector of coefficients capturing changes in effort inside the high seas pockets relative to the last year when fishing was allowed (2009). Then, μ_i are fixed-effects by grid cell and ϵ_{it} is the error term, which accounts for spatio-temporal autocorrelation via Conley standard errors.

We will also estimate changes across time and relative to changes in effort in other high seas grid cells with:

$$E_{it} = \sum_{t \neq 0} \beta_t (T_t \times D_i) + \mu_i + \tau_t + \epsilon_{it} \quad (2)$$

Here, $D_i = 1$ indicates grid cell i falls within a high seas pocket and $D_i = 0$ otherwise and τ_t are fixed-effects by year. Nested models, such as $E_{it} = \beta \text{Post}_t + \mu_i + \epsilon_{it}$ and $E_{it} = \beta \text{Post}_t \times D_i + \mu_i + \tau_t + \epsilon_{it}$, will also be estimated. In this context Post_t is a dummy variable that takes the value of 1 or 0 if the observation comes from after or before the implementation period, respectively. Models using inverse hyperbolic sine-transformed outcomes will also be estimated so that coefficients can be interpreted as percentage changes.

For H²: Let y_{it} denote longline catch-per-unit-effort (CPUE) in grid cell i at time t . We estimate:

$$y_{it} = \sum_{t \neq 0} \beta_t (T_t \times D_i) + \mu_i + \tau_t + \epsilon_{it} \quad (3)$$

All regressors have the same definitions as above. The coefficient of interest, β_t , captures the change in CPUE within the high seas pockets relative to changes observed in other high seas grid cells where purse seine effort was not curtailed. A pre-post

model with the form $y_{it} = \beta \text{Post}_t \times D_i + \mu_i + \tau_t + \epsilon_{it}$ will also be estimated to test for aggregate effects. Log-linear models will also be estimated.

For H³: Let y_{it} denote longline or purse seine catch-per-unit-effort (CPUE) in grid cell i at time t . We estimate:

$$y_{it} = \sum_{t \neq 0} \beta_t (T_t \times D_i) + \mu_i + \tau_t + \epsilon_{it} \quad (4)$$

The coefficient of interest, β_t , captures the dynamic treatment effect of the high seas closure on CPUE within 100 nautical miles of a high seas pocket, relative to changes observed in other grid cells between 100 and 200 nautical miles of a high seas pocket. A pre-post model with the form $y_{it} = \beta \text{Post}_t \times D_i + \mu_i + \tau_t + \epsilon_{it}$ will also be estimated to test for aggregate effects; log-linear models will also be estimated.

By construction, $y_{it} = \frac{\sum_{j \in (i,t)} \text{Catch}_j}{\sum_{j \in (i,t)} \text{Effort}_j}$, so $\text{Var}(y_{it}) \propto \frac{1}{\text{Effort}_{it}}$. We therefore assume

$$\text{Var}(\epsilon_{it} | X) = \frac{\sigma^2}{\omega_{it}}, \quad \omega_{it} = \text{Effort}_{it},$$

and estimate all models using normalized measures (*i.e.* CPUE) via weighted least squares using effort as weights. The measure of effort used as weight will be the same as the measure of effort used in the denominator in the CPUE calculation, which varies by gear type as stated in section 3: Dependent variable.

6 Outliers and Exclusions

Describe exactly how outliers will be defined and handled, and your precise rule(s) for excluding observations.

We will not remove any data on the basis of values reported. All data complying with the descriptions of our “Conditions” (see Section 4) and reported for years 2000-2020 will be retained.

7 Sample Size

How many observations will be collected or what will determine sample size?

We do not determine sample size, as these data are provided by the WCPFC. All data complying with the descriptions of our “Conditions” (see Section 4) will be retained.

8 Other

Anything else you would like to pre-register?

We will be using data made publicly available by the Western and Central Pacific Fisheries Commission. We will download data from the WCPFC Public Domain Aggregated Catch/Effort data download page. Specifically, Purse seine data come from: *Aggregated data, grouped by 1°x1° latitude/longitude grids, YEAR and MONTH* (WCPFC_S_PUBLIC_BY_1x1_MM5). Longline data come from: *Aggregated data, grouped by 5°x5° latitude/longitude grids, YEAR and MONTH* (WCPFC_L_PUBLIC_BY_YR_MON_3). A PDF version of this preregistration is also available.

9 Name

Give a title for this AsPredicted pre-registration

High seas pocket closures

10 Type of study.

This is an observational study

11 Data source

Data are available from the Western and Central Pacific Fisheries Commission at www.wcpfc.int/wcpfc-public-domain-aggregated-catcheffort-data-download-page